AMS 310 Final Exam Winter 2024

Survey of Probability and Statistics (Stony Brook University)

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AMS 310 Winter 2024 **Prof. Rispoli**
Final Exam

Academic integrity is expected of all students at all times, whether in the presence or absence of members of the faculty. Understanding this, I declare that I shall not give, use, or receive unauthorized aid in this examination. I have been warned that any suspected instance of academic dishonesty will be reported to the Academic Judiciary and that I will be suspended to the maximum possible penalty under university guidelines.

Name _____ ID Number _____

There are 10 problems. Each is worth 10 points. Show work for each problem.

All papers must be submitted no later than 8:30 PM. Papers submitted later than 8:30 PM will not be counted.

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Calculators are required for the test. You may not use your cell phone.

1. The random variables X and Y have the following joint probability distribution.

	f(x,y)	Y = 1	2	3	4
X	1	.1	.05	.04	.01
	2	.2	.05	.08	.04
	3	.10	.04	.10	.06
	4	.05	.00	0.01	.07

a) Find E(X)

b) Find P(Y > 2 | X = 1). Express your answer as a fraction reduced to lowest terms.

c) Find P(X > 2 | Y = 1). Express your answer as a fraction reduced to lowest terms.

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2. Continuous random variables and X and Y have a joint distribution with density function

$$f(x,y) = x^2 + (xy)/3, \text{ for } 0 < x < 1,\ 0 < y < 2$$

a) Find the conditional pdf $f(x \mid y)$.

b) Find $P(x > \frac{1}{2} \mid y = 1)$ expressed as a fraction in reduced terms.

3a) The mean and standard deviation of the heights of American women are 63.8 inches and 4 inches, respectively. Find the probability that the mean height of a randomly selected 100 American women is between 62.5 and 64.5 inches (round off to fourth decimal place).

b) Find the 80th percentile of $\overline{X}$ the mean height of a sample of 100 American women (round off to nearest whole number.)

4. a) A 90% confidence interval for a mean is going to be constructed based on a sample size of 10. What value is needed from the t distribution?

b) Find the value given by the t distribution for degrees of freedom of 15 that leaves an area of 0.025 to the left.

c) Find the value given by the Chi-Square distribution for degrees of freedom of 10 that leaves an area of 0.05 to the right.

d) Find the value given by the Chi-Square distribution for degrees of freedom of 15 that leaves an area of 0.01 to the left.

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5. The survival rate of a certain type of cancer using an existing medication is known to be 40%. A pharmaceutical company claims that the survival rate of a new drug is higher. The new drug is given to 20 patients to test this claim. Let X be the number of cures out of the 20 patients. Suppose the rejection region is $\{X \geq 10\}$. The testing hypotheses are:

H_0: $p \leq 40\%$ H_1: $p > 40\%$

Determine the type of error that can occur when the true survival rate is 45%, and calculate the error probability.

Type 2 error, and the error probability is $P(X \leq 9)$ where X is binomial, $n = 20$ and $p = 0.45$

6. A study wishes to estimate the proportion of lung cancer deaths among all cancer deaths. How large a sample of cancer death records must be examined to be 99% certain that the estimate does not differ from the true proportion by more than 0.05?

a) The American Cancer Society believes that the true proportion is 0.27. Use this information to find the required sample size.

b) Find the sample size when there is no prior knowledge about the value of p.

7. A computer company claims that batteries in a laptop last 4 hours on average. A consumer report firm gathered a sample of 16 batteries and conducted tests on this claim. The sample mean was 3 hours 50 minutes, and the sample standard deviation was 20 minutes. Assume that the battery time is distributed as normal. Carry the test to see if the battery time is significantly shorter than 4 hours at $\alpha = 0.05$. Use the 5-step method, but you do not need to find the p-value.

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8. A manufacturer claims that the standard deviation of a certain type of process is greater than 25 minutes. A sample of fifteen test panels produced a standard deviation of 31 minutes. Test the claim at $\alpha = 0.10$. List the 5 steps. You do not have to give the p-value.

9. A credit card watchdog group claims that there is a difference in the mean credit card debts of households in New York and Texas. The results of a random survey of 250 households from each state are shown below.

New York	Texas
$\bar{x}_1 = \$4446.25$	$\bar{x}_2 = \$4567.25$
$s_1 = \$1045.70$	$s_2 = \$1,361.95$
$n_1 = 250$	$n_2 = 250$

The two samples are independent. Do the results support the group's claim? Use $\alpha = 0.05$.

10. Incidence rates of asthma from samples obtained from smokers and nonsmokers are given below:

Smoker	m = 993	x = 209
Nonsmoker	n = 1012	y = 172

Test if smokers have a higher incidence rate of asthma than nonsmokers using $\alpha = 0.05$.

Identify the hypotheses, the test statistic, the critical value and the conclusion.

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Formulas and Tables for confidence Intervals and Sample Sizes

Chapter 8 Sample size needed to attain maximum error n = $\left(\frac{z_{\frac{\alpha}{2}}\sigma}{E}\right)^2$,

Sample size with β at μ = μ' for a one sided test: n = ¿¿, for a two-sided ¿¿

Normal population with known σ. A 100(1-α)% CI: ¿, $\bar{x}+z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}$)

For hypothesis testing z = $\frac{\bar{x}-\mu_0}{\frac{\sigma}{\sqrt{n}}}$. These can also be used for large samples with unknown σ, just use s instead of σ.

Small samples with unknown σ. A 100(1-α)% CI: ¿, $\bar{x}+t_{\frac{\alpha}{2},n-1}\frac{s}{\sqrt{n}}$)

Sample sizes for proportions, n = p(1-p)$\left(\frac{z_{\frac{\alpha}{2}}}{E}\right)^2$ when an estimate of p is known.

Use n = $\frac{1}{4}\left(\frac{z_{\frac{\alpha}{2}}}{E}\right)^2$ if no estimation of p is known.

A 100(1-α)% CI for p: ($\hat{p}-z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p}+z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$) where $\hat{p}=\frac{X}{n}$

A 100(1-α)% CI for variance: ($\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2},n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2},n-1}}$)

From Chapter 9

Large independent samples. A 100(1-α)% confidence interval:

$\left((\bar{x}-\bar{y})-z_{\frac{\alpha}{2}}\sqrt{\frac{s_1^2}{m}+\frac{s_2^2}{n}}, (\bar{x}-\bar{y})+z_{\frac{\alpha}{2}}\sqrt{\frac{s_1^2}{m}+\frac{s_2^2}{n}}\right)$ For hypothesis testing z = $\frac{\overline{(x¿-\bar{y})-\Delta}}{\sqrt{\frac{s_1^2}{m}+\frac{s_2^2}{n}}}$ ¿.

Small independent samples, when $\sigma_1 = \sigma_2$. A 100(1-α)% confidence interval:

$\left((\bar{x}-\bar{y})-t_{\frac{\alpha}{2},m+n-2}s_p\sqrt{\frac{1}{m}+\frac{1}{n}}, (\bar{x}-\bar{y})+t_{\frac{\alpha}{2},m+n-2}s_p\sqrt{\frac{1}{m}+\frac{1}{n}}\right)$ Where $s_p^2=\frac{(m-1)s_1^2+(n-1)s_2^2}{m+n-2}$

Small independent samples, when $\sigma_1 \neq \sigma_2$, A 100(1-α)% confidence interval:

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$$\bar{x}_1-\bar{x}_2\pm t_{\frac{\alpha}{2},v}\sqrt{\frac{s_1^2}{m}+\frac{s_2^2}{n}} \qquad \text{where } v=\frac{\left(\frac{s_1^2}{m}+\frac{s_2^2}{n}\right)^2}{\frac{\left(s_1^2/m\right)^2}{m-1}+\frac{\left(s_2^2/n\right)^2}{n-1}} \text{ (round down)}$$

For paired data. A $100(1-\alpha)\%$ confidence interval for the mean difference δ:

$$\left(\bar{d}-t_{\frac{\alpha}{2},n-1}\frac{s_d}{\sqrt{n}},\bar{d}+t_{\alpha/2,n-1}\frac{s_d}{\sqrt{n}}\right) \text{ for } n<30$$

For proportions. A $100(1-\alpha)\%$ CI for p_1-p_2 is

$$\left(\hat{p}_1-\hat{p}_2-z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{m}+\frac{\hat{p}_2(1-\hat{p}_2)}{n}},\hat{p}_1-\hat{p}_2+z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{m}+\frac{\hat{p}_2(1-\hat{p}_2)}{n}}\right), \text{ where}$$

$\hat{p}_1=x/m$ re and $\hat{p}_2=y/n$.

The Normal Distribution Table Part 1

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z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

z	Area
−3.50	0.00023263
−4.00	0.00003167
−4.50	0.00000340
−5.00	0.00000029

Source: Computed by M. Longnecker using Splus.

The Normal Distribution Table Part 2

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z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

z	Area
3.50	0.99976737
4.00	0.99996833
4.50	0.99999660
5.00	0.99999971

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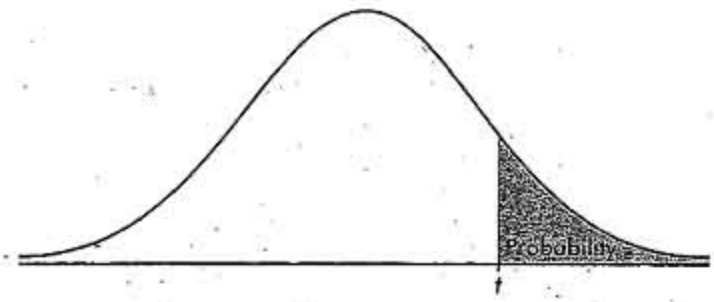

TABLE B: *t*-DISTRIBUTION CRITICAL VALUES

	Tail probability *p*											
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	.686	.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	.685	.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	.685	.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	.684	.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	.684	.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	.684	.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	.683	.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	.683	.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	.683	.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	.681	.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	.679	.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	.679	.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	.678	.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	.677	.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	.675	.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
∞	.674	.841	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.091	3.291
	50%	60%	70%	80%	90%	95%	96%	98%	99%	99.5%	99.8%	99.9%
	Confidence level *C*											

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Percentage Points of the Chi-Square Distribution

Degrees of Freedom	Probability of a larger value of x^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09
6	0.872	1.635	2.204	3.455	5.348	7.84	10.64	12.59	16.81
7	1.239	2.167	2.833	4.255	6.346	9.04	12.02	14.07	18.48
8	1.647	2.733	3.490	5.071	7.344	10.22	13.36	15.51	20.09
9	2.088	3.325	4.168	5.899	8.343	11.39	14.68	16.92	21.67
10	2.558	3.940	4.865	6.737	9.342	12.55	15.99	18.31	23.21
11	3.053	4.575	5.578	7.584	10.341	13.70	17.28	19.68	24.72
12	3.571	5.226	6.304	8.438	11.340	14.85	18.55	21.03	26.22
13	4.107	5.892	7.042	9.299	12.340	15.98	19.81	22.36	27.69
14	4.660	6.571	7.790	10.165	13.339	17.12	21.06	23.68	29.14
15	5.229	7.261	8.547	11.037	14.339	18.25	22.31	25.00	30.58
16	5.812	7.962	9.312	11.912	15.338	19.37	23.54	26.30	32.00
17	6.408	8.672	10.085	12.792	16.338	20.49	24.77	27.59	33.41
18	7.015	9.390	10.865	13.675	17.338	21.60	25.99	28.87	34.80
19	7.633	10.117	11.651	14.562	18.338	22.72	27.20	30.14	36.19
20	8.260	10.851	12.443	15.452	19.337	23.83	28.41	31.41	37.57
22	9.542	12.338	14.041	17.240	21.337	26.04	30.81	33.92	40.29
24	10.856	13.848	15.659	19.037	23.337	28.24	33.20	36.42	42.98
26	12.198	15.379	17.292	20.843	25.336	30.43	35.56	38.89	45.64
28	13.565	16.928	18.939	22.657	27.336	32.62	37.92	41.34	48.28
30	14.953	18.493	20.599	24.478	29.336	34.80	40.26	43.77	50.89
40	22.164	26.509	29.051	33.660	39.335	45.62	51.80	55.76	63.69
50	27.707	34.764	37.689	42.942	49.335	56.33	63.17	67.50	76.15
60	37.485	43.188	46.459	52.294	59.335	66.98	74.40	79.08	88.38

One Sample Tests about μ for a normal population with known σ

Step 1	$H_0: \mu = \mu_0,\ H_1: \mu \neq \mu_0$ (2-sided alternative)	$H_0: \mu = \mu_0,\ H_1: \mu > \mu_0$ (1-sided alternative)	$H_0: \mu = \mu_0,\ H_1: \mu < \mu_0$ (1-sided alternative)
Step 3	$z = \dfrac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$ Rejection Region: ¿ $z \vee \geq z_{\alpha/2}$	$z = \dfrac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$ Rejection Region: $z \geq z_{\alpha}$	$z = \dfrac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$ Rejection Region: $z \leq -z_{\alpha}$
Step 4	Substitute $\bar{x}$, σ and n Calculate z, Decision	Substitute $\bar{x}$, σ and n Calculate z, Decision	Substitute $\bar{x}$, σ and n Calculate z, Decision
Step 5	$p = 2P$ ¿	$p = P(Z \geq z)$	$p = P(Z \leq z)$

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One Sample Tests about μ for a normal population with small sample and unknown σ

Step 1	$H_0: \mu = \mu_0,\ H_1: \mu \neq \mu_0$ (2-sided alternative)	$H_0: \mu = \mu_0,\ H_1: \mu > \mu_0$ (1-sided alternative)	$H_0: \mu = \mu_0,\ H_1: \mu < \mu_0$ (1-sided alternative)
Step 3	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$
	Rejection Region: ¿ $t \vee \geq t_{\alpha/2, n-1}$	Rejection Region: $t \geq t_{\alpha, n-1}$	Rejection Region: $t \leq -t_{\alpha, n-1}$
Step 4	Substitute $\bar{x}$, s and n Calculate t, Decision	Substitute $\bar{x}$, s and n Calculate t, Decision	Substitute $\bar{x}$, s and n Calculate t, Decision
Step 5	$p = 2P$ ¿	$p = P(T \geq t)$	$p = P(T \leq t)$

One sample test for variance

Step 1	$H_0: \sigma^2 = \sigma_0^2,\ H_1: \sigma^2 \neq \sigma_0^2$ (2-sided alternative)	$H_0: \sigma^2 = \sigma_0^2,\ H_1: \sigma^2 > \sigma_0^2$ (1-sided alternative)	$H_0: \sigma^2 = \sigma_0^2,\ H_1: \sigma^2 < \sigma_0^2$ (1-sided alternative)
Step 3	$\chi^2 = \dfrac{(n-1)s^2}{\sigma_0^2}$ Rejection Region: $\chi^2 \geq \chi^2_{\alpha/2, n-1}$ or $\chi^2 \leq \chi^2_{1-\alpha/2, n-1}$	$\chi^2 = \dfrac{(n-1)s^2}{\sigma_0^2}$ Rejection Region: $\chi^2 \geq \chi^2_{\alpha, n-1}$	$\chi^2 = \dfrac{(n-1)s^2}{\sigma_0^2}$ Rejection Region: $\chi^2 \leq \chi^2_{1-\alpha, n-1}$
Step 4	Substitute s^2 and n Calculate χ^2, Decision	Substitute s^2 and n Calculate χ^2, Decision	Substitute $s^2$and n Calculate χ^2, Decision
Step 5	$p = 2min\{P(X \geq \chi^2),$	$p = P(X \geq \chi^2)$	$p = P(X \leq \chi^2)$

Two Sample Tests about μ for a normal population with large samples

Step 1	$H_0: \mu_1 - \mu_2 = \Delta_0$, $H_1: \mu_1 - \mu_2 \neq \Delta_0$	$H_0: \mu_1 - \mu_2 = \Delta_0$, $H_1: \mu_1 - \mu_2 > \Delta_0$	$H_0: \mu_1 - \mu_2 = \Delta_0$, $H_1: \mu_1 - \mu_2 < \Delta_0$
Step 3	$z = \dfrac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\dfrac{s_1^2}{m} + \dfrac{s_2^2}{n}}}$ Rejection Region: ¿ $z \vee \geq z_{\alpha/2}$	$z = \dfrac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\dfrac{s_1^2}{m} + \dfrac{s_2^2}{n}}}$ Rejection Region: $z \geq z_\alpha$	$z = \dfrac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\dfrac{s_1^2}{m} + \dfrac{s_2^2}{n}}}$ Rejection Region: $z \leq -z_\alpha$
Step 4	Substitute $\bar{x}, \bar{y}$, Calculate z, Decision	Substitute $\bar{x}, \bar{y}$, Calculate z, Decision	Substitute $\bar{x}, \bar{y}$, Calculate z, Decision
Step 5	$p = 2P$ ¿	$p = P(Z \geq z)$	$p = P(Z \leq z)$

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Two Small Independent Samples, Equal Variances

Step 1	$H_0: \mu_1 - \mu_2 = \Delta_0$, $H_1: \mu_1 - \mu_2 \neq \Delta_0$	$H_0: \mu_1 - \mu_2 = \Delta_0$, $H_1: \mu_1 - \mu_2 > \Delta_0$	$H_0: \mu_1 - \mu_2 = \Delta_0$, $H_1: \mu_1 - \mu_2 < \Delta_0$
Step 3	$t = \dfrac{(\bar{x} - \bar{y}) - \Delta_0}{s_p \sqrt{\dfrac{1}{m} + \dfrac{1}{n}}}$ Rejection Region: $\lvert t \rvert \geq t_{\frac{\alpha}{2}, m+n-2}$	$t = \dfrac{(\bar{x} - \bar{y}) - \Delta_0}{s_p \sqrt{\dfrac{1}{m} + \dfrac{1}{n}}}$ Rejection Region: $t \geq t_{\alpha, m+n-2}$	$t = \dfrac{(\bar{x} - \bar{y}) - \Delta_0}{s_p \sqrt{\dfrac{1}{m} + \dfrac{1}{n}}}$ Rejection Region: $t \leq -t_{\alpha, m+n-2}$
Step 4	Substitute $\bar{x}, \bar{y}$, s_p, m and n Calculate t, Decision	Substitute $\bar{x}, \bar{y}$, s_p, m and n Calculate t, Decision	Substitute $\bar{x}, \bar{y}$, s_p, m and n Calculate t, Decision
Step 5	$p = 2P$¿	$p = P(T \geq t)$	$p = P(T \leq t)$

Two Sample Proportion

Step 1	$H_0: p_1 - p_2 = 0$, $H_1: p_1 - p_2 \neq 0$	$H_0: p_1 - p_2 = 0$, $H_1: p_1 - p_2 > 0$	$H_0: p_1 - p_2 = 0$, $H_1: p_1 - p_2 < 0$
Step 3	$z = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\dfrac{1}{m} + \dfrac{1}{n}\right)}}$ Rejection Region: $\lvert z \rvert \geq z_{\frac{\alpha}{2}}$	$z = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\dfrac{1}{m} + \dfrac{1}{n}\right)}}$ Rejection Region: $z \geq z_\alpha$	$z = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\dfrac{1}{m} + \dfrac{1}{n}\right)}}$ Rejection Region: $z \leq -z_\alpha$
Step 4	Substitute $\hat{p}_1, \hat{p}_2$, $\hat{p}$, m and n Calculate z, Decision	Substitute $\hat{p}_1, \hat{p}_2$, $\hat{p}$, m and n Calculate z, Decision	Substitute $\hat{p}_1, \hat{p}_2$, $\hat{p}$, m, n Calculate z, Decision
Step 5	$p = 2P$¿	$p = P(Z \geq z)$	$p = P(Z \leq z)$

Probability of a Type II Error for μ

Alternative Hypothesis	$\beta(\mu')$ for a level α test
$H_1: \mu > \mu_0$	$\Phi\left(z_\alpha + \dfrac{\mu_0 - \mu'}{\sigma / \sqrt{n}}\right)$
$H_1: \mu < \mu_0$	$1 - \Phi\left(-z_\alpha + \dfrac{\mu_0 - \mu'}{\sigma / \sqrt{n}}\right)$

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$H_1: \mu \neq \mu_0$	$\Phi\left(z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right) - \Phi\left(-z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right)$

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